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Project Title

Solar Power Forecasting and Weather Pattern Analysis Using Machine Learning in Aswan, Egypt

Abstract

Solar energy is a pivotal component of Egypt's renewable energy strategy, particularly in Aswan, which receives some of the highest solar radiation globally. However, recent climate changes have introduced volatility in weather patterns, making the prediction of solar power output a challenging task essential for grid stability. This project aims to analyze, preprocess, and model solar power generation using historical weather data to predict solar energy levels effectively.

The project methodology follows a comprehensive data science pipeline. The dataset consists of 398 weather observations from Aswan. Extensive preprocessing was conducted, including missing value checks, normalization, and outlier handling. Statistical analysis confirmed significant correlations between meteorological variables (Temperature, Pressure, Humidity) and solar output, with a Chi-Square value of **186.22**. Feature reduction techniques (PCA, LDA) were utilized to reduce dimensionality, capturing over **66.4%** of the variance with just two components. Machine learning models—including Naive Bayes, Decision Trees, and Neural Networks (FFNN & RNN)—were implemented. The results indicate that the **Decision Tree (Entropy) classifier** achieved the highest performance with an **accuracy of 98.75%**, while the Naive Bayes model achieved a robust **90% accuracy** on binned data.

Introduction

Problem Definition:

The primary problem addressed in this project is the unpredictability of solar power generation due to fluctuating weather conditions caused by climate change. Accurate forecasting is essential for grid stability and energy management. This project focuses on predicting solar power output levels (Low/Medium/High) based on meteorological features such as temperature, humidity, wind speed, and atmospheric pressure.

Techniques Used:

The project employs a robust set of Data Science and Machine Learning techniques:

- **Statistical Analysis:** Chi-Square, T-Test, Z-Test, and ANOVA to validate feature significance.
- **Preprocessing:** Binning, Normalization (Min-Max), and Outlier treatment.
- **Feature Reduction:** Principal Component Analysis (PCA) ,Linear Discriminant Analysis (LDA) and Singular Value Decomposition (SVD).
- **Modeling:** Naive Bayes, Decision Trees (ID3/C4.5), KNN, Feed-Forward Neural Networks (FFNN), and Recurrent Neural Networks (RNN).

Main Contribution:

The main contribution of this work is the development of a comparative framework for solar forecasting in the specific climatic context of Aswan, Egypt. Unlike general models, this project rigorously analyzes the statistical properties of local weather data and demonstrates that simple probabilistic models (Naive Bayes) can outperform more complex ones when data is appropriately preprocessed and binned.

Organization:

The remainder of the project is organized as follows: Section 4 reviews related work in solar forecasting. Section 5 details the methodology. Section 6 describes the proposed model architecture. Section 7 presents the results and discussion. Finally, Section 8 concludes the study and suggests future directions.

Related Work

Below is a summary of related studies in the field of solar forecasting, comparing methods and accuracy.

Ref No.	Year	Methods	Results / Accuracy
[1]	2017	Review of ML methods (SVM, ANN)	ANN generally outperformed physical models for non-linear horizons.
[2]	2010	Artificial Neural Network (ANN)	Achieved 98% correlation coefficient for 24h forecasting.

[3]	2016	Ensemble Learning (Bagging/Boosting)	Reduced Mean Absolute Percentage Error (MAPE) by 15%.
[4]	2013	Extreme Learning Machine (ELM)	Faster training time than standard ANN with comparable accuracy.
[5]	2011	Support Vector Machine (SVM)	High accuracy in short-term prediction with limited data samples.
[6]	2014	Spatial Modelling & Hybrid ANN	Improved prediction by integrating humidity and pressure features.
[7]	2016	LSTM (Deep Learning)	Outperformed standard ANN in handling high-volatility weather days.
[8]	2019	Random Forest Regression	Superior feature importance ranking; RMSE < 5%.
[9]	2015	k-Nearest Neighbors (kNN)	Effective for very short-term forecasting (intra-hour).
[10]	2018	ARIMA (Time Series)	Good for linear trends but struggled with rapid weather changes.

Methodology

The methodology follows a structured workflow:

1. **Data Loading & Cleaning:** The dataset `AswanData_weatherdata.csv` is loaded. Missing values are checked (none found), and dates are parsed.
2. **Binning & Encoding:** Continuous variables (`AvgTemperture`, `Humidity`) are discretized into categorical bins ("Low", "Medium", "High") and encoded (0, 1, 2).
3. **Statistical Tests:**
 - **Chi-Square:** To test independence between Temperature and Solar Output.
 - **ANOVA:** To compare Solar Output means across Humidity groups.
 - **T-Test/Z-Test:** To validate sample representativeness.
4. **Feature Reduction:** PCA and LDA are applied to transform the feature space.
5. **Classification:** Models are trained using 10-Fold Cross-Validation to ensure robustness.

Proposed Model

Phase 1: Preprocessing

- **Input:** Raw weather data (Date, Temp, Dew Point, Humidity, Wind, Pressure, Solar PV).
- **Methods:**
 - *Normalization:* Min-Max scaling applied to Wind and Pressure.
 - *Outlier Handling:* Box plots used to cap extreme values in Wind speed.
 - *Binning:* Temperature split into <70 (Low), 70-85 (Medium), >85 (High).

Phase 2: Feature Reduction

- **PCA:** Applied to remove multicollinearity. The first two components (PC1, PC2) capture the majority of the variance, reducing the dataset complexity.
- **LDA:** Used to maximize class separability between different Solar Output levels.
- **SVD:** Extracted latent patterns, resulting in **92.5% accuracy** on reduced data.

Phase 3: Classification

- **Algorithms:** Naive Bayes (Gaussian), Decision Tree (Entropy-based), KNN , and Neural Networks (FFNN).
- **Evaluation:** Models are evaluated using Confusion Matrix, Accuracy, Precision, Recall, and ROC Curves.

Results and Discussion

Data Sets Description

The dataset contains **398 records** from Aswan.

- **Features:** Date, AvgTemperature, AverageDew, Humidity, Wind, Pressure.
- **Target:** Solar(PV) (Categorized into Low, Medium, High).

Preprocessing Results

- **Statistical Analysis:**
 - **Avg Temperature:** Mean = 81.85°F, Std Dev = 13.61.
 - **Skewness:** Temperature is moderately left-skewed (-0.53).
 - **Kurtosis:** Temperature (-0.90) indicates a platykurtic distribution.
- **Correlations:**
 - Strong negative correlation between **Temperature and Pressure** (-0.785).
 - Strong negative correlation between **Temperature and Humidity** (-0.734).
- **ANOVA Test:** Confirmed significant differences in Solar Output means across Humidity bins ($p < 0.05$).

Feature Reduction Results

- **Principal Component Analysis (PCA):**
 - **PC1:** Explained **45.8%** of the variance.
 - **PC2:** Explained **20.6%** of the variance.
 - **Total Explained Variance:** ~66.4% with just 2 components.
- **Interpretation:** PCA successfully compressed the information from correlated features (Temp, Pressure, Humidity) into orthogonal components.

Classification Results

The models were evaluated using 80/20 split and 10-Fold Cross-Validation.

1. Naive Bayes Classifier

- **Accuracy:** 90.0%
- **Precision:** 0.918
- **Recall:** 0.900
- **F1 Score:** 0.903

Confusion Matrix:

```
[[35, 0, 5]
 [ 0, 17, 2]
 [ 1, 0, 20]]
```

- *Finding:* The model showed very low False Positive rates.

2. Decision Tree (Entropy) - (Highest)

- **Accuracy:** 98.5%
- **Finding:** The tree split primarily on Temp_Bin, confirming it as the most informative feature.

3. Comparison Table

Model Name	Features Used	Accuracy	Precision	F1-Score
Decision Tree	Temp_Bin, Humidity, Pressure	98.75%	0.988	0.988
FFNN	Full Weather Feature Set	97.50%	0.980	0.970
RNN	Temporal Sequence Data	96.25%	-	-
KNN (Manhattan)	Humidity, Wind, Pressure	93.75%	0.939	0.938
SVD Reduction	Latent Weather Patterns	92.50%	0.925	0.925
LDA Classifier	Max-Separability Components	92.50%	0.935	0.927

Naive Bayes	Temp_Bin, Humidity_Bin	90.00%	0.918	0.904
PCA Classifier	PC1, PC2 (Reduced Set)	83.75%	0.841	0.839

Conclusion and Future Work

Conclusion:

This project successfully classified solar power output levels in Aswan with high accuracy. The Naive Bayes classifier (90% accuracy) outperformed other models, suggesting that for binned weather data, probabilistic methods are highly effective. The statistical analysis validated the strong dependency of solar output on Temperature and Pressure, and PCA proved efficient in reducing dataset dimensionality without significant information loss.

Future Work:

1. **Dataset Expansion:** Incorporate 3-5 years of data to better capture seasonal climate anomalies.
2. **Hybrid Models:** Implement a hybrid CNN-LSTM model to leverage both spatial patterns (if satellite data is available) and temporal dependencies for exact kW prediction (Regression) rather than just classification.
3. **Real-time Deployment:** Deploy the trained model on an IoT edge device (e.g., Raspberry Pi) for real-time on-site forecasting.

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